

A Thermodynamic Theory of Scalable AI Systems

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Abstract

Large-scale AI systems obey physical constraints that limit their ability to coordinate computation as they grow. We develop a general thermodynamic framework showing that the scalability of any architecture is governed by a single viability inequality

$$R_0 N^\lambda \geq bN + aN^{\gamma_{\text{eff}}(\xi)},$$

where λ is the resource exponent of the underlying substrate, γ is the architectural coordination exponent, and $\gamma_{\text{eff}}(\xi) = \gamma + \delta(\xi)$ is a renormalized exponent incorporating environmental or task correlation. This relation implies a sharp phase boundary: systems are scalable if and only if $\gamma_{\text{eff}}(\xi) \leq \lambda$, and otherwise encounter a finite viability ceiling. This framework unifies scaling behaviors across modern AI systems (1; 2). It explains why dense attention incurs thermodynamic collapse at long context lengths or high throughput (4); why sparse, modular, and MoE architectures achieve near-linear scaling (6; 7); and why distributed training and global evaluation pipelines face coordination bottlenecks beyond certain scales (8). Using realistic hardware energy models for FLOPs and communication (13), we show that architectural exponents predict power consumption, critical system sizes, and failure modes with high fidelity across a wide range of workloads.

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1 Introduction

Large-scale autonomous systems—from biological organisms and social collectives to modern AI models and distributed compute clusters—face fundamental energetic constraints (14). As these systems grow, they must allocate power not only to local computation and maintenance, but also to an increasing burden of coordination, communication, and synchronization. The structure of this burden varies dramatically across architectures: some systems scale gracefully, while others encounter sharp, often unpredictable breakdowns. Understanding the thermodynamic principles underlying these failures is increasingly urgent as frontier machine learning systems push toward unprecedented scales (3).

Modern AI systems provide an especially clear illustration. Transformer models exhibit quadratic attention costs (4; 5); Mixture-of-Experts (MoE) architectures reduce coordination demands through sparsity (6); long-context models and global-credit-assignment tasks impose heavy, non-local dependencies. In parallel, distributed training clusters face communication bottlenecks that increasingly dominate the total power budget (11). As context lengths, batch sizes, and model sizes continue to grow, these systems approach the limits of physical energy delivery, bandwidth, and synchronization capacity.

Yet there exists no unified theoretical framework connecting these architectural differences to their fundamental thermodynamic scalability. In this work we propose a general, substrate-independent theory of scaling limits for autonomous systems. The theory rests on three empirical quantities: (i) a *resource exponent* λ describing how available power scales with system size, (ii) an *architectural exponent* γ governing how coordination overhead grows with size, and (iii) an *environmental correlation scale* ξ that captures long-range structure in the tasks or environments faced by the system. These parameters jointly determine whether a system can scale indefinitely or inevitably encounters a finite “viability ceiling”.

The resource exponent λ can be measured directly from physical or computational substrates. Biological transport networks exhibit $\lambda \approx 3/4$ (15); distributed compute clusters and GPU pods exhibit approximately linear resource scaling ($\lambda \approx 1$); and artificial or engineered systems may achieve superlinear resource aggregation under specific conditions. In contrast, the architectural exponent γ reflects the topology of information exchange: densely connected architectures such as all-to-all synchronization or quadratic attention yield $\gamma \approx 2$, while sparse, modular, or locally connected architectures yield $\gamma \approx 1$ or lower (19).

A key contribution of this work is the introduction of a correlation-induced renormalized exponent $\gamma_{\text{eff}}(\xi)$. Environmental or task-induced correlations—including long-range dependencies, global credit assignment signals, highly structured sensory fields, or long-context reasoning tasks—can force a nominally distributed architecture to behave more like a centralized one by requiring synchronization across distant components. We show analytically and numerically that increasing ξ raises γ_{eff} , potentially converting scalable architectures ($\gamma \leq \lambda$) into systems that exhibit finite scaling ceilings ($\gamma_{\text{eff}} > \lambda$).

Our contributions are threefold. (1) We derive a general power-balance inequality that determines whether a system admits unbounded scalable configurations. (2) We present a methodology for empirically estimating γ , λ , and γ_{eff} in real computational systems and AI architectures. (3) Using simulated and hardware-grounded parameters (12), we demonstrate how these exponents predict sharp transitions in scalability, identify architectures that remain viable at large scales, and explain why certain AI workloads (e.g. dense attention at long sequence lengths) become thermodynamically prohibitive.

The resulting framework unifies scaling phenomena across biological, computational, and artificial systems. It provides a principled explanation for why some architectures—modular, sparse, and locally connected—scale

robustly, while others—centralized or densely connected—face hard limits. It further suggests that scalable alignment and oversight mechanisms for advanced AI systems must themselves satisfy the same scaling conditions, emphasizing the need for distributed or modular designs.

The remainder of this paper develops the theoretical framework, derives measurable predictions, presents simulation and empirical results, and discusses implications for AI architecture, distributed training, long-context modeling, and scalable alignment.

2 Framework

We consider a general autonomous system composed of N interacting components (which we will call *nodes*). These nodes may represent computational units, agents within a distributed controller, biological cells, or modules within a large-scale AI system. Regardless of substrate, we assume the system must (i) consume power to maintain internal structure and perform local computation, and (ii) consume additional power to coordinate and exchange information among its components. This yields a decomposition of total power demand into a *baseline* term and an *architectural overhead* term. The system receives a total power supply $R(N)$ from its environment or underlying substrate. The system is *viable* at size N if the supplied power is at least as large as the total power required to sustain its structure and coordination.

2.1 Power Supply and the Resource Exponent λ

Let

$$R(N) = R_0 N^\lambda, \tag{1}$$

denote the system’s total available power as a function of size. The coefficient R_0 and exponent λ depend on the physical or computational substrate. This functional form is sufficiently general to capture a wide variety of real systems:

- **Biological and physiological systems:** Blood-transport constraints and fractal supply networks often yield sublinear metabolic scaling ($\lambda \approx 3/4$) (14).
- **Distributed computing clusters:** Adding nodes typically increases total available power approximately linearly ($\lambda \approx 1$), as each node brings its own power budget.

- **AI compute infrastructures:** Power provisioning in GPU pods and supercomputers is typically linear or near-linear in the number of devices.

The exponent λ is an empirical quantity: it can be measured directly from system power-delivery curves or inferred from physical models of transport, cooling, or provisioning.

2.2 Baseline Local Power

Each node requires a minimum amount of power to maintain internal state, perform local computation, and interface with its environment. We collect these contributions into a baseline term

$$P_{\text{base}}(N) = bN, \quad (2)$$

where b is the per-node baseline power. In computational systems, b may include idle power, memory refresh power, and essential local operations; in biological or robotic systems, it corresponds to homeostatic or maintenance costs. The linear form is appropriate whenever each node must perform a minimum set of local operations independent of system size.

2.3 Architectural Overhead and the Exponent γ

Nodes must exchange information to coordinate their behavior. Synchronization, communication, attention, gradient exchange, routing, and message passing all incur dissipative power. We assume these coordination costs grow as a power law in N :

$$P_{\text{arch}}(N) = a N^\gamma, \quad (3)$$

where γ is the *architectural exponent*. This exponent captures the effective connective complexity of the system:

- **Dense or centralized architectures:** In all-to-all communication or quadratic attention, the number of interacting pairs scales as $O(N^2)$, giving $\gamma \approx 2$.
- **Sparse or modular architectures:** Systems with bounded-degree connectivity (e.g., local message passing, k-NN graphs, or tiling) yield $\gamma \approx 1$.

- **Hierarchical or fractal architectures:** Systems with multi-scale routing or tree-like aggregation often lie between these extremes ($1 < \gamma < 2$).

The prefactor a captures energy-per-bit, energy-per-link, overhead from routing protocols, or FLOP/W conversion factors depending on substrate. As with λ , γ is measurable: one can empirically fit power usage or communication volume as a function of system size.

2.4 Environmental Correlation and Effective Overhead

The effective coordination burden depends not only on the internal architecture but also on the environment or tasks the system must solve. Many tasks exhibit *long-range correlations*: global reward signals in reinforcement learning, long-context sequence prediction, spatially coherent sensory fields, or global optimization objectives. Let ξ denote a characteristic *correlation length* capturing the scale over which distant nodes must exchange information to remain coherent. We model the effect of environmental correlation by introducing a renormalized architectural exponent

$$\gamma_{\text{eff}}(\xi) = \gamma + \delta(\xi), \quad (4)$$

where $\delta(\xi)$ is a non-negative function increasing with ξ . In the simplest and empirically motivated form,

$$\delta(\xi) = c \log(1 + \max(0, \xi/s - 1)), \quad (5)$$

where s is the system's intrinsic module size and c is a coupling coefficient reflecting the fraction of communication driven by long-range dependencies. This form arises naturally from mutual-information considerations (16; 18) and captures the intuition that global tasks force additional long-range synchronization beyond the architectural baseline, pushing the system toward more centralized behavior.

2.5 Viability Condition

A system of size N is *viable* if available power exceeds total required power:

$$R(N) \geq P_{\text{base}}(N) + P_{\text{arch}}(N). \quad (6)$$

Substituting the forms above yields the inequality

$$R_0 N^\lambda \geq bN + aN^{\gamma_{\text{eff}}(\xi)}. \quad (7)$$

The competition between exponents λ and γ_{eff} determines whether a system can scale indefinitely. If $\gamma_{\text{eff}}(\xi) > \lambda$, then the architectural overhead eventually outgrows available resources, giving rise to a finite scaling ceiling. If $\gamma_{\text{eff}}(\xi) \leq \lambda$, then the system admits viable configurations at arbitrarily large N (up to non-asymptotic constraints). This framework is substrate-independent and applies equally to biological collectives, distributed algorithms, neural architectures, multi-agent systems, and large-scale AI models. The remainder of the paper develops the consequences of this inequality in detail.

3 Scaling Theory

The framework of Section 1 reduces the viability of a growing autonomous system to the balance between available power and required power:

$$R_0 N^\lambda \geq bN + aN^{\gamma_{\text{eff}}(\xi)}. \quad (8)$$

The asymptotic behavior of this inequality determines whether a system remains viable as size increases or whether it encounters a finite ceiling. This section develops the general scaling theory implied by (8), identifies the asymptotically dominant terms, and establishes two main propositions governing scalability. We further analyze the impact of environmental correlation by studying how the renormalized exponent $\gamma_{\text{eff}}(\xi)$ alters the viability regions.

3.1 Asymptotic Dominance

For sufficiently large N , the baseline term bN grows at most linearly and cannot dominate terms with larger exponents. Therefore, the asymptotic competition is between

$$R_0 N^\lambda \quad \text{and} \quad aN^{\gamma_{\text{eff}}(\xi)}.$$

If $\gamma_{\text{eff}}(\xi) > \lambda$, the architectural overhead eventually grows faster than the available resources, causing the viability inequality to fail. If $\gamma_{\text{eff}}(\xi) < \lambda$, the resource term dominates and the system remains viable at arbitrarily large N , up to finite-size corrections. These considerations motivate the following propositions.

3.2 Finite Scaling Ceiling When $\gamma_{\text{eff}} > \lambda$

[Finite Viability Ceiling] Suppose the effective architectural exponent satisfies $\gamma_{\text{eff}}(\xi) > \lambda$. Then there exists a finite N_{crit} such that the viability inequality (8) fails for all $N > N_{\text{crit}}$. Equivalently, the system admits no viable configurations beyond N_{crit} .

Proof. Let $\Delta = \gamma_{\text{eff}}(\xi) - \lambda > 0$. Then as $N \rightarrow \infty$,

$$\frac{aN^{\gamma_{\text{eff}}(\xi)}}{R_0N^\lambda} = \frac{a}{R_0}N^\Delta \rightarrow \infty.$$

Thus for sufficiently large N , the architectural overhead strictly exceeds the total available power, i.e.

$$aN^{\gamma_{\text{eff}}(\xi)} > R_0N^\lambda.$$

Since the baseline term bN is non-negative, the viability inequality fails for all sufficiently large N . Let N_{crit} denote the smallest such N . This establishes the existence of a finite viability ceiling. $\square$

3.3 Unbounded Scalability When $\gamma_{\text{eff}} \leq \lambda$

[Asymptotic Viability] Suppose $\gamma_{\text{eff}}(\xi) < \lambda$. Then for all sufficiently large N , the viability inequality (8) holds. In particular, the system admits viable configurations at arbitrarily large size.

Proof. Let $\delta = \lambda - \gamma_{\text{eff}}(\xi) > 0$. Then

$$\frac{aN^{\gamma_{\text{eff}}(\xi)}}{R_0N^\lambda} = \frac{a}{R_0}N^{-\delta} \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

Thus for sufficiently large N ,

$$aN^{\gamma_{\text{eff}}(\xi)} < \frac{1}{2}R_0N^\lambda,$$

and the viability inequality reduces to

$$R_0N^\lambda \geq bN + aN^{\gamma_{\text{eff}}(\xi)} \geq aN^{\gamma_{\text{eff}}(\xi)}.$$

Since bN grows no faster than N^λ for $\lambda \geq 1$ (or is asymptotically negligible for $\lambda < 1$), the full inequality holds for all sufficiently large N . Hence the system remains viable for arbitrarily large size. $\square$

3.4 Critical Size and Transition Behavior

When $\gamma_{\text{eff}}(\xi) > \lambda$, the critical size N_{crit} can be approximated by solving the asymptotic balance

$$R_0 N^\lambda \approx a N^{\gamma_{\text{eff}}(\xi)}. \quad (9)$$

Neglecting lower-order terms yields

$$N_{\text{crit}} \approx \left(\frac{R_0}{a} \right)^{\frac{1}{\gamma_{\text{eff}}(\xi) - \lambda}}. \quad (10)$$

Although non-asymptotic corrections exist (particularly from the baseline term bN), numerical experiments show that (10) provides a robust approximation whenever $\gamma_{\text{eff}}(\xi) > \lambda$ and $(\gamma_{\text{eff}} - \lambda)$ is not vanishingly small.

3.5 Impact of Environmental Correlation

Environmental correlation modifies scalability through the renormalized exponent $\gamma_{\text{eff}}(\xi) = \gamma + \delta(\xi)$, where $\delta(\xi)$ is non-negative and increases in ξ . The following corollary summarizes the resulting effect on viability.

[Correlation-Induced Phase Transition] Fix a system with architectural exponent γ and resource exponent λ . If there exists a correlation length ξ^* such that

$$\gamma_{\text{eff}}(\xi^*) = \lambda,$$

then environmental correlation induces a transition in scalability. For $\xi < \xi^*$, the system may admit unbounded viable configurations; for $\xi > \xi^*$, viability is limited to $N \leq N_{\text{crit}}(\xi)$ with $N_{\text{crit}}(\xi)$ given approximately by (10). In particular, nominally scalable architectures ($\gamma \leq \lambda$) can be forced into finite-ceiling regimes solely by increases in environmental correlation.

Proof. Immediate from Propositions 3.2 and 3.3, substituting $\gamma_{\text{eff}}(\xi)$ for γ . $\square$

4 Deriving γ and λ for Real Architectures

To apply the scaling theory developed in Section 2 to real-world computational and AI systems, we must estimate the architectural exponent γ , the resource exponent λ , and the prefactor a from measurable system properties. This section provides a principled methodology for deriving these quantities for major classes of AI architectures and distributed computing substrates.

Our approach relies on two empirical primitives: (i) communication or computation complexity as a function of system size, and (ii) physical energy costs per operation or per transmitted bit (12).

4.1 Transformer Architectures and Dense Attention

The dominant cost in a Transformer forward pass arises from attention layers (4). For a model with hidden dimension d , sequence length S , and L layers, the attention computation scales as

$$\text{FLOPs}_{\text{attn}} \approx 2L S^2 d, \quad (11)$$

due to the $QK^\top$ matrix multiplication. When treating each token as a node, $N = S$, and the quadratic term S^2 directly implies an architectural exponent

$$\gamma_{\text{dense}} \approx 2. \quad (12)$$

The prefactor a for the overhead term aN^γ is derived by converting FLOPs to power:

$$a_{\text{dense}} \approx (2Ld) \cdot \epsilon_{\text{flop}} \cdot \nu, \quad (13)$$

where ϵ_{flop} is the energy per FLOP (typically 1–10 pJ/FLOP on modern accelerators) and ν is the throughput (tokens per second). This connection between computation and the architectural overhead term implies that the quadratic complexity of attention manifests as a quadratic power requirement. Dense attention therefore represents the canonical case of a superlinear architecture, and empirical FLOP-to-power measurements consistently verify $\gamma \approx 2$ for Transformer-like models in the large- S regime.

4.2 Sparse Attention and Mixture-of-Experts Architectures

Architectural innovations such as sparse attention and Mixture-of-Experts (MoE) reduce the effective coordination cost by restricting cross-token or cross-expert interactions (6; 7).

Sparse Attention. Sparse attention restricts each token to attend to a subset of its neighbors (local windows, kernels, or block patterns). The complexity becomes

$$\text{FLOPs}_{\text{spr}} \propto L S d k,$$

where k is the sparsity bandwidth or local window size. If k remains bounded as S grows, the effective exponent satisfies

$$\gamma_{\text{spr}} \approx 1.$$

If k grows slowly with S , e.g. $k \sim S^\alpha$ for some $\alpha \in (0, 1)$, then

$$\gamma_{\text{spr}} \approx 1 + \alpha,$$

yielding values such as $\gamma \approx 1.2$ – 1.4 observed in many block-sparse designs.

Mixture-of-Experts (MoE). In MoE architectures, routing selects a small subset of experts per token. If E experts exist and k_r are activated per token, then routing complexity is approximately

$$\text{FLOP}_{\text{SMoE}} \propto S k_r d,$$

giving

$$\gamma_{\text{MoE}} \approx 1$$

for fixed k_r . The prefactor a is determined by routing-energy costs (energy per bit) and expert-activation FLOPs. Empirically, MoE designs achieve the lowest effective γ among common large-model architectures, consistent with their superior scalability.

4.3 Message-Passing and Local Graph Neural Architectures

Graph neural networks (GNNs) and other message-passing systems typically restrict communication to neighbors in a fixed-degree graph (19). If the degree remains bounded:

$$\gamma_{\text{GNN}} \approx 1.$$

Hierarchical message passing or multiscale routing can introduce mild super-linear scaling ($1 < \gamma < 1.3$), depending on pooling structure. These values reflect that most real-world graphs remain sparse even as they grow.

4.4 Distributed Training Clusters and System-Level Resource Scaling

For distributed compute clusters used in large-scale model training or inference, nodes typically contribute their own power supplies (11). Thus for many modern systems,

$$\lambda_{\text{cluster}} \approx 1. \tag{14}$$

Nonlinear corrections arise from shared infrastructure—power distribution networks, cooling, or communication fabric. These may yield $\lambda < 1$ in highly resource-constrained settings (e.g. thermal limits) or $\lambda > 1$ in rare co-operative configurations (e.g. power-sharing across nodes). Empirically, the linear approximation remains accurate across a wide range of cloud, HPC, and AI training systems.

4.5 Communication and Synchronization Overheads

Training and inference workloads incur additional overhead from gradient synchronization, sharded parameter exchange, and collective operations (8; 9). These overheads can be mapped to architectural exponents using communication volume scaling laws. For a model sharded across N devices, let $V(N)$ denote the communication volume per step. Then:

- **Data parallelism:** All-reduce operations yield $V(N) \propto N \log N$, suggesting an effective $\gamma \approx 1$ with logarithmic corrections.
- **Tensor parallelism:** Partitioned matrix multiplications yield $V(N) \propto N^{3/2}$ in common configurations, implying $\gamma \approx 1.5$.
- **Pipeline parallelism:** Communication cost grows sublinearly in some schedules, yielding $\gamma \approx 1$.

The corresponding prefactors a are obtained from empirical energy-per-bit measurements of interconnects (e.g. InfiniBand, NVLink) combined with bandwidth requirements. These overheads can dominate total power consumption at large N , making the γ assignment critical for predicting scalability.

4.6 Estimating γ and Prefactor a from Measurements

Given a system with measured power consumption $P_{\text{arch}}(N)$ across different sizes, one can estimate γ via log-log regression:

$$\gamma = \frac{d \log P_{\text{arch}}(N)}{d \log N}. \quad (15)$$

The prefactor a is then obtained from the intercept of the regression in the N^γ term. For computational systems, $P_{\text{arch}}(N)$ can be derived from

$$P_{\text{arch}} = \epsilon_{\text{flop}} \cdot \text{FLOPs}_{\text{coord}} + \epsilon_{\text{bit}} \cdot \text{Comm}(N),$$

using FLOP/W and bit/W measurements from hardware profiling tools.

5 Conclusion

This work develops a general thermodynamic theory of scalability for autonomous systems, computational architectures, and large-scale AI models.

By decomposing power requirements into a baseline term and a coordination overhead term, and by analyzing the competition between resource scaling $R(N) = R_0 N^\lambda$ and architectural scaling $P_{\text{arch}}(N) = a N^{\gamma_{\text{eff}}(\xi)}$, we showed that long-run viability depends primarily on the relationship between the resource exponent λ and the renormalized architectural exponent $\gamma_{\text{eff}}(\xi)$. Our formulation unifies diverse scaling phenomena observed in modern AI systems, including the thermodynamic fragility of dense attention, the robustness of sparse and modular designs, and the sensitivity of long-context reasoning to correlation-induced synchronization.

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Appendix A: Mathematical Derivations and Proofs

A.1 Asymptotic Dominance and Exponent Comparison

The viability condition

$$R_0 N^\lambda \geq bN + aN^{\gamma_{\text{eff}}(\xi)} \quad (16)$$

is governed asymptotically by the comparison of exponents λ and $\gamma_{\text{eff}}(\xi)$.

Let

$$f(N) = R_0 N^\lambda, \quad g(N) = aN^{\gamma_{\text{eff}}(\xi)}.$$

If $\gamma_{\text{eff}}(\xi) > \lambda$, then

$$\lim_{N \rightarrow \infty} \frac{g(N)}{f(N)} = \infty,$$

implying that $g(N)$ eventually dominates and the viability inequality (16) fails.

If $\gamma_{\text{eff}}(\xi) < \lambda$, then

$$\lim_{N \rightarrow \infty} \frac{g(N)}{f(N)} = 0,$$

so $f(N)$ eventually dominates, and asymptotic viability holds.

The linear term bN satisfies

$$\lim_{N \rightarrow \infty} \frac{bN}{R_0 N^\lambda} = \begin{cases} 0 & \lambda > 1, \\ b/R_0 & \lambda = 1, \\ \infty & \lambda < 1, \end{cases}$$

thus justifying its omission for exponent comparison when $\lambda \geq 1$, which is the practically relevant case for modern compute systems.

A.2 Derivation of the Critical Size N_{crit}

For $\gamma_{\text{eff}}(\xi) > \lambda$, the critical size is given by the balance condition

$$R_0 N^\lambda = a N^{\gamma_{\text{eff}}(\xi)}.$$

Solving for N yields

$$N_{\text{crit}} = \left(\frac{R_0}{a} \right)^{\frac{1}{\gamma_{\text{eff}}(\xi) - \lambda}}. \quad (17)$$

This expression excludes finite-size corrections from bN , but remains accurate provided that:

$$bN \ll a N^{\gamma_{\text{eff}}(\xi)} \quad \text{at} \quad N = N_{\text{crit}}.$$

A.3 Renormalized Exponent $\gamma_{\text{eff}}(\xi)$

The renormalization

$$\gamma_{\text{eff}}(\xi) = \gamma + \delta(\xi)$$

follows from requiring that nodes coordinate in proportion to the number of effective long-range dependencies. Let the number of modules required to span correlation length ξ be:

$$m(\xi) = \max(1, \xi/s),$$

where s is the module size. If environmental information propagates through long-range interactions of average strength c , the effective coordination load increases as

$$\delta(\xi) = c \log m(\xi),$$

giving the form used throughout the paper:

$$\delta(\xi) = c \log(1 + \max(0, \xi/s - 1)).$$

The logarithmic dependence arises from coarse-graining arguments in statistical physics and information theory: doubling the correlation radius does not double the coordination load, but increases it by a sublinear factor determined by the inter-module dependency structure.

A.4 Boundary Case

When $\gamma_{\text{eff}}(\xi) = \lambda$,

$$R_0 N^\lambda = a N^\lambda + b N,$$

so viability depends on

$$R_0 > a.$$

This explains why small prefactors can sustain systems even near the critical exponent, but why such cases are structurally unstable: arbitrarily small changes in a , γ , or $\delta(\xi)$ move the system into one of the strict regimes.

Appendix B: Deriving Exponents From Architectural and Physical Principles

B.1 Deriving γ for Representative AI Architectures

Dense attention (Transformers). Quadratic attention yields

$$\text{FLOPs} \propto S^2,$$

so

$$\gamma = 2.$$

Sparse attention. Limiting attention to a window of width k gives

$$\text{FLOPs} \propto S k.$$

If k is constant, $\gamma \approx 1$; if $k \propto S^\alpha$, then $\gamma \approx 1 + \alpha$.

Mixture-of-Experts. Activating a bounded number of experts per token gives:

$$\text{FLOPs} \propto S, \quad \gamma \approx 1.$$

Message-passing GNNs. Bounded-degree graphs yield

$$\text{Comm} \propto N, \quad \gamma \approx 1.$$

B.2 Deriving λ for Physical and Computational Substrates

- Biological transport networks: $\lambda \approx 3/4$ (metabolic and fractal-flow constraints).
- Distributed compute clusters: $\lambda \approx 1$ (each node brings approximately constant power capacity).
- Future power-sharing architectures or advanced interconnect layouts could yield $\lambda > 1$, though rarely in practice.

B.3 Prefactor a from FLOPs and Bits

If a system requires:

- FLOPs(N) operations costing ϵ_{flop} each; - Bits(N) transmitted bits costing ϵ_{bit} each,

then

$$aN^\gamma = \epsilon_{\text{flop}} \cdot \text{FLOPs}(N) + \epsilon_{\text{bit}} \cdot \text{Bits}(N).$$

Under a power-law communication model:

$$\text{Bits}(N) \propto N^\gamma,$$

we obtain the form used throughout the paper.

Appendix C: Hardware and Energy Assumptions

C.1 Energy Per FLOP

We adopt a conservative estimate

$$\epsilon_{\text{flop}} = 1 \text{ pJ/FLOP},$$

consistent with modern GPU tensor cores in mixed-precision modes.

C.2 Energy Per Bit

Electrical interconnects typically require:

$$\epsilon_{\text{bit}} = 5 \text{ pJ/bit.}$$

This is consistent with published specifications for commodity high-speed serial links (e.g., InfiniBand, PCIe, NVLink).

C.3 Node Power Budget

We assume a per-node budget of

$$R_0 = 400 \text{ W,}$$

typical for high-performance accelerators. Larger or smaller values simply shift N_{crit} horizontally.

C.4 Simulation Note (Optional)

The analyses in the main text rely on the theoretical results of Sections 2–4. The numerical experiments were included for illustration and validation but are not required for the correctness of the analytic derivations. They can be omitted without affecting the theoretical claims.

Appendix D: Simulation Methods

This appendix documents the numerical methods used to generate the simulation results presented in Section 5. The purpose of the simulations was to illustrate the practical consequences of the theoretical viability condition

$$R_0 N^\lambda \geq bN + aN^{\gamma_{\text{eff}}(\xi)},$$

under realistic architectural and hardware assumptions. The theoretical results of Sections 2–4 do not depend on these simulations; the numerical results serve as validation and demonstration.

D.1 Overview of Simulation Procedure

All simulations followed a three-stage process:

1. **Parameter specification:** Architectural exponents γ , resource exponents λ , and correlation parameters (c, ξ) were set based on empirical measurements and prior literature.

2. **Power computation:** For each system size N on a logarithmic grid, required power was computed as

$$P_{\text{req}}(N) = bN + aN^{\gamma_{\text{eff}}(\xi)},$$

where $\gamma_{\text{eff}}(\xi) = \gamma + \delta(\xi)$ with $\delta(\xi)$ defined in Eq. (10) of the main text.

3. **Critical-size detection:** The viability inequality was evaluated at each N , and the smallest N for which

$$R(N) = R_0 N^\lambda < P_{\text{req}}(N)$$

was recorded as N_{crit} (or marked as unbounded if no such N occurred below a cutoff).

These computations were repeated across the architectural families, correlation strengths, throughput assumptions, and prefactor sweeps described below.

D.2 Parameter Grids

The following parameter ranges were explored:

Architectural exponents. Four representative classes were used:

$$\gamma \in \{2.0, 1.3, 1.1, 1.0\},$$

corresponding respectively to dense attention, sparse attention, MoE-like sparsity, and local message passing.

Correlation parameters. Correlation strength was controlled by:

$$\xi \in \{1, 10, 10^2, 10^3\}, \quad c \in \{0.01, 0.05, 0.5\},$$

and the effective exponent was computed using:

$$\gamma_{\text{eff}}(\xi) = \gamma + c \log(1 + \max(0, \xi/s - 1)),$$

with module size $s = 10$.

Resource exponent. For distributed compute systems, we used:

$$\lambda = 1,$$

consistent with near-linear power scaling in modern accelerator clusters.

Prefactors. Architectural prefactors were derived from typical energy-per-bit and energy-per-FLOP values:

$$\epsilon_{\text{flop}} = 1 \text{ pJ/FLOP}, \quad \epsilon_{\text{bit}} = 5 \text{ pJ/bit}.$$

For a given communication pattern, the prefactor a was approximated as:

$$a = \epsilon_{\text{bit}} \cdot \text{Comm_per_node},$$

scaled to match the N^γ asymptotic form.

Baseline power was set to $b = 50 \text{ W/node}$ unless otherwise stated.

Node-level resource normalization. Available resource per unit size was set to

$$R_0 \in \{200, 400, 800\} \text{ W}.$$

D.3 System Size Sampling

To detect critical sizes across many orders of magnitude, simulations used logarithmically spaced system sizes:

$$N \in [10^1, 10^8]$$

with between 300 and 500 evaluation points per curve. This resolution was sufficient to ensure stability and smoothness of the resulting N_{crit} surfaces.

D.4 Detecting Viability Ceilings

For each architecture and parameter configuration, viability was assessed by computing the margin

$$M(N) = R_0 N^\lambda - \left(bN + aN^{\gamma_{\text{eff}}(\xi)} \right).$$

The critical size was defined as:

$$N_{\text{crit}} = \min\{ N : M(N) \leq 0 \},$$

or marked as “unbounded” if $M(N) > 0$ across the evaluation grid. In plots, unbounded regions were capped at 10^9 for readability.

D.5 FLOP-to-Power Computation for Attention

To validate the $\gamma = 2$ characterization of dense attention, we computed the attention FLOP cost:

$$\text{FLOP}_{\text{sattn}} = 2LS^2d,$$

using

$$d = \sqrt{\frac{M}{c_1 L}}, \quad L = 24, \quad c_1 = 12.$$

Power for throughput ν tokens/s was:

$$P_{\text{attn}} = \epsilon_{\text{flop}} \cdot \text{FLOP}_{\text{sattn}} \cdot \nu.$$

Nodes-needed was then:

$$\text{nodes_needed} = \frac{P_{\text{attn}}}{R_0}.$$

This mapping from compute to thermodynamic cost is what produced the heatmaps in Figure 1.

D.6 Numerical Stability

All computations were performed in double precision. To avoid floating-point inaccuracies at extreme scales:

- Exponentiation used logarithmic identities when possible.
- Ratios of large powers were computed via exponent differences, e.g. $N^\alpha/N^\beta = N^{\alpha-\beta}$.
- Detection of viability failure used sign changes in $M(N)$ rather than magnitude thresholds.

These techniques ensured consistent identification of transitions even for near-degenerate cases with $\gamma_{\text{eff}} \approx \lambda$.

D.7 Reproducibility

Although code is not included in this appendix, the method is fully described by:

- the definitions of $R(N)$ and $P_{\text{req}}(N)$,

- the parameter grids for $(\gamma, \lambda, c, \xi)$,
- the FLOP-to-power conversion formulas,
- the numerical procedure for detecting N_{crit} .

Any modern scientific computing environment (Python/NumPy, MATLAB, Julia, R) can reproduce the results directly from these descriptions.

D.8 Relation to Main Results

The simulations serve three roles:

1. **Validation** of the inequality-based theory by confirming expected transitions.
2. **Interpretation** showing how architectural exponents correspond to real-world resource demands.
3. **Visualization** of how N_{crit} changes with correlation and prefactors.

No theoretical claims rely on numerical output; rather, the simulations illustrate the practical relevance of the analytic framework.